REQUIREMENTS GATHERING

Stock Market Analytics & Investment Decision Support Dashboard

Project Information	Details
Program	DEPI – Professional Data Analysis Track
Phase	Requirements Gathering
Team Name	[Stock Vision]
Team Members	Member 1: [أحمد عبدالحميد محمد الجندي] Member 2: [أحمد عبدالباسط مصطفى] Member 3: [رقية محمد عبدالسلام شلبي] Member 4: [زينب عبدالمنعم دياب] Member 5: [بيشوي سامي يوسف بقطر] Member 6: [محمد إبراهيم عبدالله غنيم]
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1. Purpose and Project Overview

This document identifies the intended users, stakeholder needs, system requirements, and business questions for a dashboard that analyzes historical stock-market data obtained from Yahoo Finance, subject to the source's available fields, access conditions, and permitted use.

The dashboard is intended to help users explore price movements, compare selected companies, understand historical performance and variability, and review indicators that may support further research. It is an analytical and educational decision-support tool—not a guarantee of future returns or personalized financial advice.

2. Project Objectives

- Collect historical daily price data for a team-approved list of publicly traded companies.
- Prepare and validate the dataset, including date formats, missing values, duplicate records, and invalid prices.
- Calculate clear, reproducible performance indicators from the available data.
- Build an interactive dashboard for company comparison, time-period selection, and trend exploration.
- Document data-source limitations, update frequency, and the meaning of each metric.

3. Scope

3.1 In Scope

- Historical daily Open, High, Low, Close, and Volume fields where available for the selected tickers.
- Exploratory analysis of price trends, daily changes, percentage returns, trading range, and historical volatility where data supports the calculation.
- Company and date-range filters, comparative visuals, KPI cards, and a concise summary of findings.

3.2 Out of Scope

- Guaranteed predictions of future stock prices or investment returns.
- Personalized buy/sell instructions or financial advice.
- Execution of trades, brokerage account integration, or portfolio transactions.
- Claims of real-time or intraday updates unless the data source and access method are verified to provide them.

4. Stakeholder Analysis

Stakeholder	Main Need / Interest	Expected Involvement
Project team / data analysts	Understand data, clean it, define metrics, build visuals, and validate results.	High — requirements, development, testing, documentation.
Dashboard users / learners	Explore historical performance and compare selected companies easily.	Medium — provide feedback on usability and useful views.
Project supervisor / DEPI reviewer	Assess whether the project is structured, transparent, and meets the stated scope.	Medium — review deliverables and provide feedback.
Data source (Yahoo Finance)	Provides market data subject to its available fields, access conditions, and terms.	External dependency — verify access and usage conditions.
Potential investor / researcher	Use historical indicators as one input to independent research.	Potential user — no assumption that the dashboard gives personalized advice.

5. User Stories and Use Cases

ID	User Story	Acceptance / Success Condition
US-01	As a dashboard user, I want to select a company ticker so I can inspect its historical price behavior.	Selecting a valid ticker updates the relevant visuals and summary metrics.
US-02	As a user, I want to choose a date range so I can focus on a specific period.	Charts and calculations reflect the selected range; invalid or empty ranges are handled clearly.
US-03	As a user, I want to compare two or more companies so I can examine differences in historical performance.	The comparison view uses consistent dates, units, and clearly labeled metrics.
US-04	As a user, I want to see daily price changes and percentage returns so I can understand movements over time.	The dashboard defines the calculation and handles missing or zero prior prices safely.
US-05	As a user, I want to see high/low ranges and a volatility indicator so I	Indicators are calculated from documented formulas and are

ID	User Story	Acceptance / Success Condition
	can explore historical variability.	labeled as historical.
US-06	As a user, I want to know when the data was last refreshed so I can judge its recency.	The dashboard or accompanying documentation displays the last successful refresh date/time.
US-07	As a reviewer, I want to trace metrics back to source fields and formulas so I can validate the analysis.	A metric dictionary or methodology section documents inputs, formulas, and limitations.

6. Functional Requirements

ID	Requirement	Priority
FR-01	Load historical daily data for an approved list of stock tickers from a verified Yahoo Finance access method.	Must
FR-02	Retain a consistent company identifier and trading date for each observation.	Must
FR-03	Validate data types and flag missing values, duplicate observations, and invalid or implausible values for review.	Must
FR-04	Calculate selected metrics such as daily absolute change and daily percentage return using documented formulas.	Must
FR-05	Show price trends over time for the selected company or companies.	Must
FR-06	Allow users to filter by ticker/company and date range.	Must
FR-07	Provide a comparison view using consistent definitions and clearly labeled units.	Should
FR-08	Display selected summary indicators, such as latest available close, period return, and historical high/low, when the required data is available.	Should
FR-09	Document data refresh date/time, source, and any failed or incomplete refresh.	Must
FR-10	Provide a metric glossary and a brief limitations/disclaimer note.	Must
FR-11	Export a filtered table or summary if supported by the selected tool and project timeline.	Could

7. Non-Functional Requirements

ID	Quality Area	Requirement
NFR-01	Usability	Labels, chart titles, filters, and units should be clear to a beginner user.
NFR-02	Accuracy	Metrics must be calculated from documented formulas and checked against sample records.
NFR-03	Data quality	Missing or invalid records must be flagged or handled using a documented rule, not silently ignored.
NFR-04	Performance	The dashboard should remain reasonably responsive for the selected dataset and project environment.
NFR-05	Reliability	Refresh failures or unavailable data should be reported rather than presented as a successful update.
NFR-06	Reproducibility	The team should record the source, retrieval date, selected tickers, transformations, and metric definitions.
NFR-07	Transparency	The dashboard must distinguish historical observations from forecasts and avoid implying guaranteed outcomes.
NFR-08	Maintainability	Data preparation and metric calculations should be organized and documented so the team can update them.

8. Business Questions

1. How have the closing prices of the selected companies changed over the selected historical period?

2. Which company achieved the highest and lowest percentage returns during the same period?

3. How do daily percentage returns differ across the selected companies?

4. Which company experienced the highest and lowest historical volatility based on the standard deviation of daily returns?

5. What were the highest and lowest recorded closing prices for each selected company during the analysis period?

6. How do trading volumes change over time, and which companies recorded the highest average daily trading volume when reliable data is available?

7. How do company performance rankings change when different historical date ranges are selected?

8. Which companies performed best and worst based on historical returns and volatility, and how does considering both measures affect the comparison?

9. Which companies experienced the largest increases and decreases in closing prices during the analysis period, and how significant were these changes?

10. Is there a relationship between trading volume and daily stock price movements among the selected companies?

9. Data Requirements and Definitions

Field / Concept	Purpose	Notes
Ticker / company identifier	Distinguishes each stock.	Use a team-approved ticker list; confirm ticker symbols and exchange context.
Date	Supports time-series analysis and date filtering.	Use consistent date parsing and account for trading days.
Open, High, Low, Close	Describe daily price behavior.	Confirm the exact fields returned by the selected access method.
Adjusted Close	May support adjusted historical comparisons where available.	Confirm availability and document whether calculations use Close or Adjusted Close.
Volume	Supports trading-activity exploration.	Use only when available and interpret its units consistently.
Daily change	Close minus prior available close.	Define how gaps and missing prior observations are handled.
Daily percentage return	(Current close / prior close − 1) × 100.	State whether Close or Adjusted Close is used; avoid division by zero.
Historical volatility	Summarizes variability in returns over a specified period.	Define the method and window; it describes the past, not future risk.

10. Assumptions, Constraints, and Risks to Requirements

- The exact ticker list and historical date range have not yet been finalized; the team must agree on them.
- Data fields, historical coverage, refresh options, access restrictions, and permitted usage must be checked against the chosen Yahoo Finance access method before implementation.
- Daily historical data should not be described as real-time or intraday data.
- Stock splits, dividends, currency differences, market holidays, and differing trading calendars can affect comparisons; the team should document how these are treated.
- Historical performance does not guarantee future results. Any ranking must be based on transparent project criteria and should not be presented as personalized investment advice.
- If data retrieval is unavailable or incomplete, the team should document the limitation and use only data it can legitimately access and validate.

11. Initial Acceptance Criteria

- The dashboard loads the agreed set of companies and displays the selected date range correctly.
- At least one time-series view and one company-comparison view are available.
- Core calculations are documented and manually checked against sample records.
- Filters update all intended visuals consistently.
- The last successful data refresh is recorded and visible in the dashboard or documentation.
- Known data-quality issues and project limitations are documented.
- The dashboard labels all performance and risk metrics as historical and includes a clear non-advisory disclaimer.

End of document